Hall Ticket Number:

MR-22

Code: B425CM5

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Regular/Supplementary Examinations, July 2025

Object Oriented Programming through Java

(Computer Science and Engineering (AI and ML))

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What is the purpose of a constructor in Java?	1	1	1	1
1. b)	Describe the use of this keyword in Java.	1	2	1	2
1. c)	Explain the concept of substitutability in inheritance.	1	2	2	1
1. d)	How does polymorphism help in object-oriented design?	1	2	2	2
1. e)	What is autoboxing in Java?	1	1	3	1
1. f)	Define daemon thread with an example.	1	2	3	1
1. g)	Why are adapter classes useful in event handling?	1	3	4	2
1. h)	Describe how a scrollbar is different from a scrollpane.	1	2	4	2
1. i)	Why is Swing preferred over AWT?	1	2	5	1
1. j)	What method adds a tab in a JTabbedPane?	1	3	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Define Object-Oriented Programming. What are the key principles of OOP?	5	1	1	1
2. b)	Explain the concept of garbage collection in Java. Why is it important?	5	2	1	2
	OR				
2. c)	Discuss the difference between method overriding and method overloading with examples.	5	2	1	2
2. d)	Illustrate the concept of parameter passing between objects in OOP.	5	2	1	2
3. a)	What is the difference between a base class and a subclass? Give an example.	5	1	2	1
3. b)	Define a Java interface *Shape* with a method *area()*. Implement it in two classes *Circle* and *Rectangle* and display their areas.	5	3	2	3
	OR				
3. c)	Explain the concept of CLASSPATH and its significance in using packages.	5	2	2	1
3. d)	Write a Java program that defines an interface Printable and implements it in a class Document to print a message.	5	3	2	3
4. a)	Describe the difference between termination and resumptive exception handling models.	5	2	3	1
4. b)	Write a Java program that creates two threads: one prints even numbers, the other prints odd numbers up to 10.	5	3	3	3
	OR				
4. c)	Explain the life cycle of a thread in Java with a neat diagram.	5	1	3	1
4. d)	Write a Java program to demonstrate the use of try, catch, and finally blocks.	5	2	3	1
5. a)	Explain the delegation event model with a suitable diagram.	5	2	4	2
5. b)	Write a Java AWT program that responds to a button click using. ActionListener.	5	3	4	3

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	OR				
5. c)	How does Java handle mouse events using event listeners?	5	2	4	2
5. d)	Compare and contrast Flow, Grid layout managers in terms of flexibility and usage.	5	3	4	2
6. a)	Describe the MVC architecture with a neat diagram.	5	2	5	1
6. b)	How are parameters passed to an applet? Illustrate with a simple program.	5	4	5	2
	OR				
6. c)	Design a simple calculator using Swing components like JTextField, JButton, and appropriate event handling.	10	5	5	3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome